DID EPRC Workshop Talk

Working Slide Outline With Planned Figures and Tables

Draft working document

October 2026

Overview

This document drafts the main content for each slide in the DID EPRC workshop talk. It is not the final slide deck. Each entry gives the slide-facing text and then shows the table, equation, or figure that would appear on that slide.

The empirical motivation is Strickland, Chan, and Gottfried (2026), an *Educational Evaluation and Policy Analysis* paper on NYC's Gifted and Talented (G&T) program using a staggered difference-in-differences design. The workshop exercises use synthetic panel data generated for teaching, inspired by that setting; they are not the NYC administrative data used in the paper.

Part A. DID Setup and Intuition

Slide 1. Title and Roadmap

Difference-in-differences (DID) is widely used in education policy and program evaluation when some students, schools, or districts are exposed to a policy and others are not. This talk starts with the simplest DID setup, then explains how the same logic becomes a TWFE model, an event-study model, and finally a staggered adoption design.

Approx. time	**Topic**
0–5 min	Why DID is useful for education policy questions.
5–15 min	Traditional DID and parallel trends.
15–25 min	Regression DID, TWFE, and event-study models.
25–35 min	Why TWFE becomes harder with staggered adoption.
35–45 min	Preview of staggered DID estimators and handoff.

Planned table for Slide 1: roadmap for the 45-minute talk.

Slide 2. Policy Example: NYC Gifted and Talented

Policy example / research question. NYC's Gifted and Talented (G&T) program. Do students who enter G&T experience better outcomes after entry than comparable students who do not enter? This example is motivated by Strickland, Chan, and Gottfried (2026), which studies the longitudinal effects of NYC's G&T program using a staggered DID design.

Important clarification. The workshop data are synthetic. They are inspired by this education policy setting, but they are not the NYC administrative data used in the paper.

DID piece	G&T example	Synthetic workshop data
Unit	Student, school, or district	District.
Time	School year or grade	Year.
Treatment	G&T entry	Treatment turns on.
Treated group	Students exposed to G&T	Districts treated in year 5.
Comparison group	Not-yet or never exposed students	Never-treated districts.
Outcome	Achievement, attendance, or behavior	Standardized achievement outcome.

Planned table for Slide 2: policy example mapped to the synthetic data.

Slide 3. Traditional DID Setup

In the simplest DID design, we compare two groups across two periods. One group receives treatment after the pre-period. The other group is not treated. DID asks whether the treated group changes more after treatment than the comparison group changes over the same period.

$$\widehat{\delta}_{DID} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre}).$$

Concept. DID is not just treated versus comparison after treatment. It is the treated group's change minus the comparison group's change.

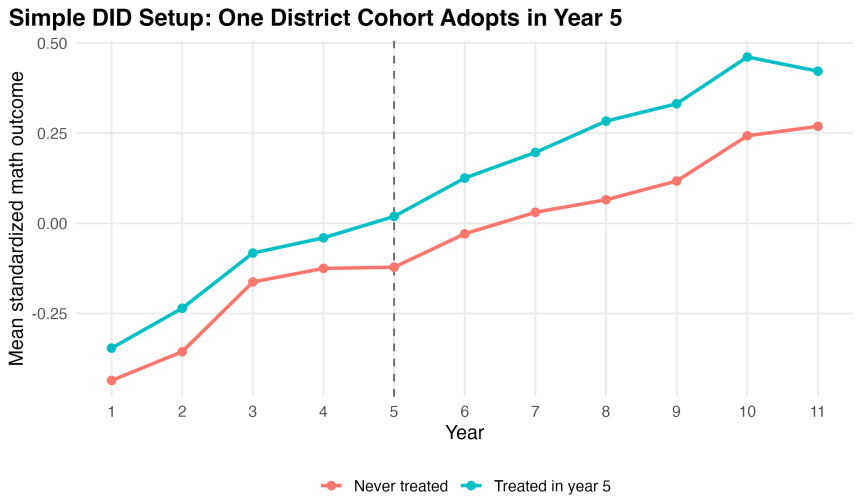
Group	Pre	Post	Change
Treated	Mean before year 5	Mean after year 5	Treated change.
Comparison	Mean before year 5	Mean after year 5	Comparison change.
DID			Treated change minus comparison change.

Planned table for Slide 3: two-group, two-period DID structure.

Slide 4. Parallel Trends: The Identifying Assumption

Key identifying idea. DID asks what would have happened to treated units after treatment if they had not been treated. The comparison group is useful if it tells us what would have happened to the treated group in the absence of treatment.

$$E[Y_{T,post}(0) - Y_{T,pre}(0)] = E[Y_{C,post}(0) - Y_{C,pre}(0)].$$



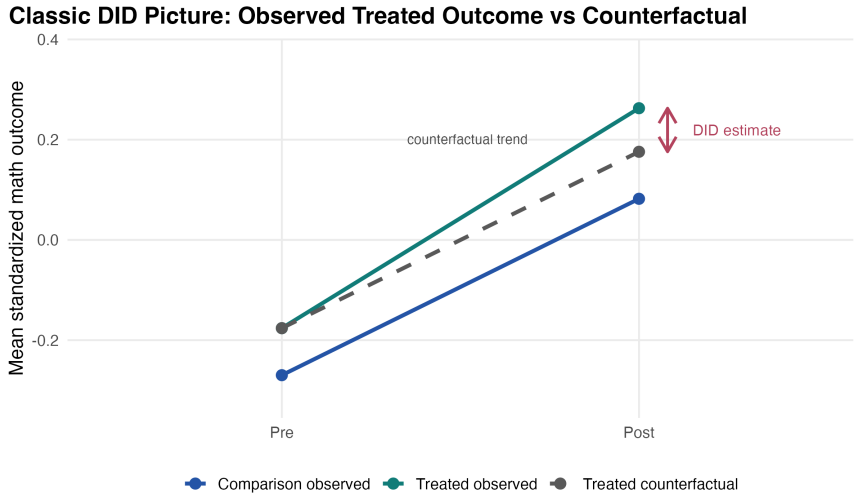
Planned figure for Slide 4: raw group trends from the synthetic data. Treated units adopt in year 5; comparison units are never treated.

Slide 5. Classic DID Picture: Observed vs. Counterfactual

This is the classic DID teaching figure. The treated group has an observed post-treatment outcome, but the missing object is the treated group's counterfactual post-treatment outcome: where would the treated group have been without treatment?

$$\widehat{Y}_{T,post}(0) = \bar{Y}_{T,pre} + (\bar{Y}_{C,post} - \bar{Y}_{C,pre}).$$

$$\widehat{\delta}_{DID} = \bar{Y}_{T,post} - \widehat{Y}_{T,post}(0).$$



Planned figure for Slide 5: comparison trend, observed treated trend, dashed treated counterfactual trend, and the DID estimate.

Slide 6. DID as a Regression

The same two-by-two contrast can be estimated with a regression. The coefficient on the interaction between treatment group and post period is the DID estimate. Because the data are generated, we can compare the estimated effect with the known simulated effect.

$$Y_{it} = \alpha + \beta Treat_i + \gamma Post_t + \delta(Treat_i \times Post_t) + \varepsilon_{it}.$$

Term	**Meaning**
$Treat_i$	Treated units differ from comparison units.
$Post_t$	Outcomes differ after treatment begins.
$Treat_i \times Post_t$	Treated units change more or less after treatment.
δ	DID estimate.

Planned table for Slide 6: terms in the two-by-two DID regression.

Model	**Estimated effect**	**Known simulated effect**
Two-by-two DID regression	0.087	0.090

Planned table for Slide 6: estimated DID effect versus the known simulated effect.

Part B. TWFE and Event Studies

Slide 7. Introduce the TWFE Model

Introduce the TWFE model here. Many applied DID papers estimate a two-way fixed effects (TWFE) model. Instead of only using one pre-period and one post-period, TWFE uses the full panel and controls for unit fixed effects and time fixed effects.

$$Y_{it} = \alpha_i + \lambda_t + \delta D_{it} + \varepsilon_{it}.$$

Concept. In the simple one-treatment-cohort case, TWFE maps closely onto the familiar DID contrast. The model is attractive because it uses repeated observations and absorbs stable unit differences and common time shocks.

Model piece	Interpretation
α_i	Unit fixed effects: stable differences across districts.
λ_t	Year fixed effects: shocks common to all districts in a year.
D_{it}	Treatment indicator that equals 1 after treatment begins.
δ	Average treatment effect in the simple setup.

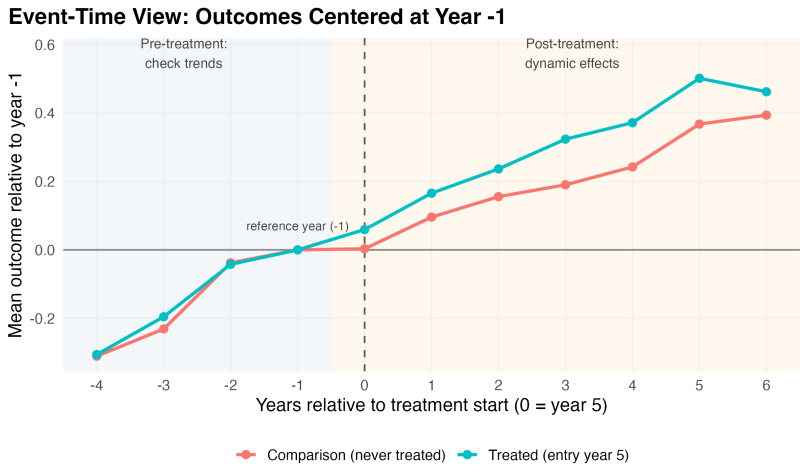
Planned table for Slide 7: pieces of the TWFE model.

Slide 8. Event Studies: Re-centering Time Around Treatment

Introduce the event-study version of TWFE here. An event study replaces the single treatment indicator with indicators for time relative to treatment. This lets us look at outcomes before and after treatment entry. In the G&T example, event time 0 is the year a student enters G&T. In the synthetic data, event time 0 is year 5.

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k 1[t - G_i = k] + \varepsilon_{it}.$$

Concept. Pre-treatment coefficients describe whether treated and comparison units were already diverging. Post-treatment coefficients show how effects evolve after treatment. Event time -1 is omitted as the reference period.



Planned figure for Slide 8: outcomes centered at event time -1. Pre-treatment event times show whether treated and comparison groups were already diverging; post-treatment event times show how effects evolve after treatment.

Slide 9. Limitation of a Single TWFE Coefficient

Explain the limitation of TWFE: one coefficient can hide treatment-effect dynamics. A single TWFE coefficient is simple, but it compresses many possible effects into one number. This can be misleading if effects grow, fade, or appear only several years after treatment.

In the simple design, TWFE is close to the familiar DID contrast. Once treatment timing varies, TWFE can compare newly treated units to never-treated units, not-yet-treated units, and already-treated units in ways that are harder to interpret.

What TWFE gives us	**What it can hide**
One treatment coefficient	Effects that change over time.
A simple average	Differences across treatment cohorts.
A familiar regression model	Comparisons involving already-treated units.

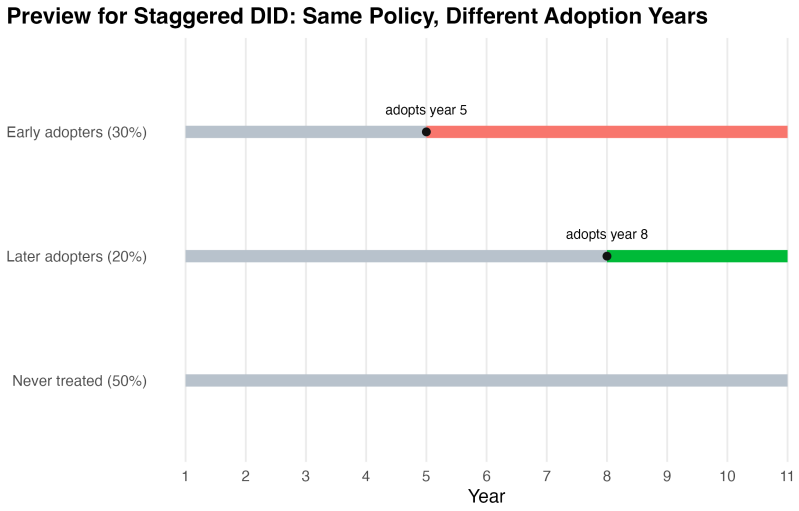
Planned table for Slide 9: why one TWFE coefficient can be too compressed.

Part C. Staggered DID Preview

Slide 10. Staggered DID Example: G&T Entry at Different Times

Use G&T to introduce staggered treatment timing. In many education settings, treatment does not start for everyone at the same time. Students may enter G&T at different grade points. Districts or schools may adopt a policy in different years.

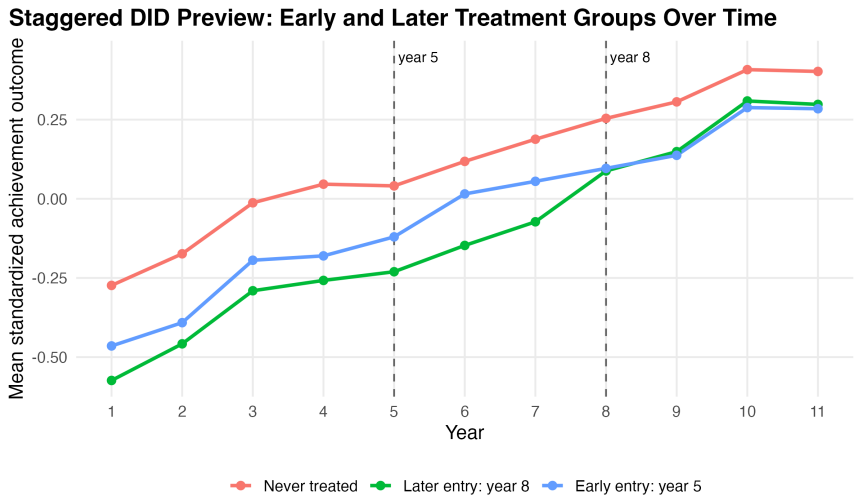
In the synthetic data, one treated group adopts in year 5 and another adopts in year 8. A third group is never treated. This mirrors the core staggered-adoption problem: treatment timing is itself part of the design.



Planned figure for Slide 10: early adopters, later adopters, and never-treated units.

Slide 11. Why Staggered Timing Complicates TWFE

Explain the problem of TWFE under staggered adoption. With staggered timing, the comparison group changes over time. In some years, later-treated units can serve as comparisons for early-treated units. After the later group becomes treated, those comparisons become more complicated.



Planned figure for Slide 11: early-treated, later-treated, and never-treated group trends.

Comparison type	Interpretation
Treated vs. never treated	Usually the most intuitive DID comparison.
Early treated vs. not-yet treated	Useful before the later group adopts.
Later treated vs. already treated	Can be problematic if effects evolve over time.

Planned table for Slide 11: why staggered timing changes the comparison problem.

This motivates the Goodman-Bacon decomposition: TWFE with staggered adoption can be understood as a weighted average of many two-by-two DID comparisons, not all of which answer the same policy question.

Slide 12. Preview of Staggered DID Estimators

Preview CSA logic: estimate group-time effects, then aggregate. Modern staggered DID estimators avoid relying on one opaque TWFE coefficient. Instead, they estimate treatment effects for specific groups at specific times.

$$ATT(g,t) = E[Y_t(g) - Y_t(0) \mid G_i = g].$$

For group g, first treated in period g, estimate the effect at time t by comparing that group to appropriate untreated or not-yet-treated units. Then aggregate group-time effects into the summary needed for the research question.

$$ATT = \sum_{g,t} w_{gt} ATT(g,t).$$

Estimand	Meaning
$ATT(g,t)$	Effect for the cohort first treated in period g at time t.
Simple aggregation	Average over group-time effects.
Group aggregation	Average effects by treatment cohort.
Event-time aggregation	Average effects by time since treatment.

Planned table for Slide 12: CSA-style group-time estimands and aggregations.

Slide 13. Brief Alternatives and Reading Group Handoff

Briefly introduce alternatives; do not teach them fully here. This workshop will not fully cover every modern staggered DID estimator. The goal is to show why we need alternatives to naive TWFE and where those alternatives fit.

Method	What it helps clarify
Goodman-Bacon decomposition	What comparisons TWFE is using.
Callaway and Sant'Anna	Group-time treatment effects.
Wooldridge pooled OLS / ETWFE	Regression-based alternatives.
Sun and Abraham	Event-study estimates with staggered timing.

Planned table for Slide 13: methods to flag before Wei's next section.

The introductory section closes by pointing participants to the workshop materials: the synthetic data, starter code, outline/slides, and readings. The next section can pick up with covariates and conditional parallel trends.

https://katjanestrickland.github.io/did-eprc-workshop/

References To Include

Strickland, K., Chan, W., and Gottfried, M. A. (2026). Longitudinal effects of NYC's Gifted and Talented program: Evidence from a staggered difference-in-differences approach. *Educational Evaluation and Policy Analysis*. https://doi.org/10.3102/01623737261438146

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254–277.

Callaway, B., and Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200–230.

Sun, L., and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2), 175–199.

Wooldridge, J. M. (2021). Two-way fixed effects, the two-way Mundlak regression, and difference-in-differences estimators.